50.012 Networks

Project opening briefing

2026 Spring

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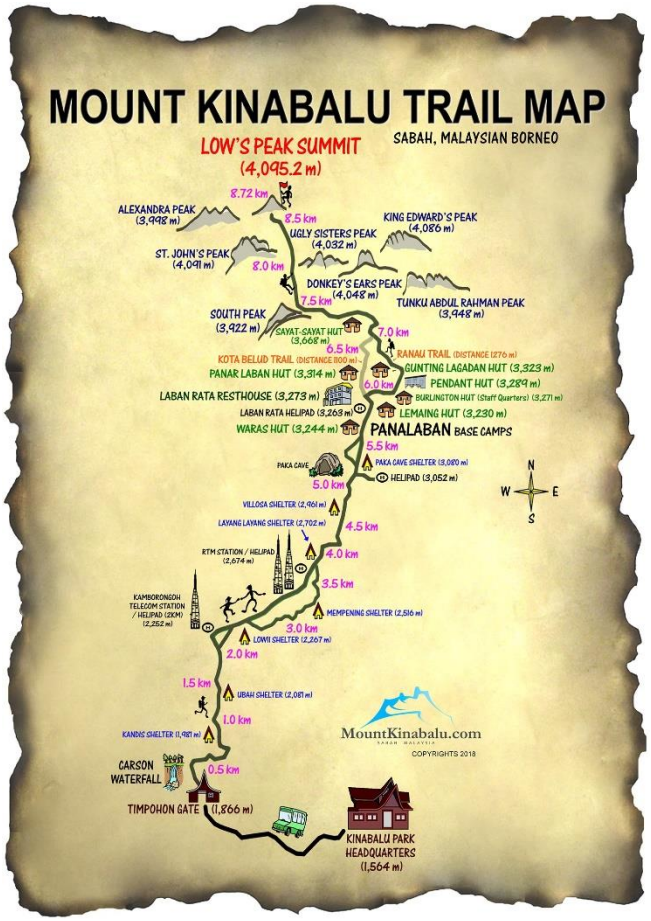
Project: logistics

- Grade: 18pt +2 bonus points
  - Main checkpoints:
    - By next week: teams formed
    - 27 March & 30 March (week 9 Friday & Week 10 Monday): Project midpoint presentation (problem statement, literature review, approach statement) What problem will you tackle? What is your plan to create something new?
    - 17 Apr & 20 Apr (Week 12 Friday & Week 13 Monday): final presentation What have you achieved in your project.
  - Week 13: final submission (report, demo code / experiment data, working journal, peer review)



Project: logistics

- 3 - 5 members / group
  - 49 students -> ~ 12 groups
  - Expect to spend ~ 30 hours / person on the project (spend them in 2-3 hours / chunk)
  - Each group has 90 - 150 hours to tackle some interesting network problem together!
    - Work as a team.
    - Project management: Task allocation. Start from a small, specific question. Keep iterating. Keep a working journal.

Project: budget

- TBC
  - I am requesting for 100 SGD budget / group
  - Will need each group to appoint a POC to receive the funding (if it is approved)

Mission *possible*

- In this project, your team will propose a ***specific*** problem related to network application / protocol / connectivity / architecture … and trek together to prototype / experiment with something new for your chosen topic

- Theme for this year: **Innovation & Entrepreneurship (I&E) in Computer Networks Field**

1	2	3	4	5	6	7	8	9	10	11	12	13
WiFi Router bandwidth allocation	Cloud Provisioning BTS	AWS	Connecting...	Video Call BTS	Livestreaming	Internet Begone!	Social Messaging Platform Traffic and DDOS	VPN	How to Roam	Internet race	How does Telegram work?	the Great Firewall
How does WiFi Routers allocate bandwidth for users within the same local network? What happens when users disconnect or when more users join the same network, what underlying mechanism(s) governs this?	How a few clicks on a Cloud Provider's web browser leads to the allocation of virtual resources which are connected, isolated, and installed with the services ready for you to use.	Understand AWS's ecosystem	Explore what happens behind the scenes when a device switches networks?	What happens behind the scenes when you make a zoom / Whatsapp video call to your friend?	How does live-streaming work and what protocols are used to manage it?	How to turn off the internet for the entire country and only give access to your friends	Given the number of users social messaging platforms such as WhatsApp, Telegram and Messenger has, how do they manage traffic and prevent DDOSing by malicious parties.	How are some VPNs able to bypass China's Great Firewall	How network service providers track your geographic location, determine when to activate roaming, etc		Network traffic of Telegram	Empirical Analysis of On-Path Injection Mechanisms in Heterogeneous Networks
Ting Kok Mun (086)	Skipper Smahon (162)	Finn Yap Jun Hun (122)	Chintakunta Shreya (937)	Ng Ding Jie (843)	Ow Yong Jia Hui (207)	Sng Wei Qi Amos (952)	Tirkey Shania Sarah (069)	Benedict Chua Kuan Hsien (200)	Richard Zhang (179)	Pranav Harwadekar	Khoo Zi Qi (984)	xulai（308）
Nguyen Thanh Nam (987)	Chua Wei Yang (828)	Koh Niann Tsyr (136)	Wong Pui Ling (172)	Emata Michael Jr Rasco (854)	Bryan How Jun Teck (189)	Chen Jing Hao (183)	Chia Tang(200)	Alaina Akriti Jaiswal (150)	Daniel Truong (195)	Eirikur Gudni Thoarinsson	Sean Ho Junxian (041)	
Ng Keng Hwang (978)	Bennet Boey (077)	Melissa Lee Wei Qing (933)	Wonna Tun (687)	Lim Zhiyu, Kendrix (152)	Ong Wei Wen. Viviana (982)	Pea Jun Hao Darren (859)	Ng Au Hern Wesley (888)	Tan Jing Yu (127)	Albert Nguyen-Tran (204)	Thorsteinn Heangur Jonsson	Koo Rou Zhen (038)	@Xu Lai, would you like to join Group 9?
Nicholas Yow Jun Hui (997)		Ong Wei Yang (126)	Xue Yufeng (834)	Fan Xiangwei (533)		-	Megha Pusti (128)	Maheshwari Nair (470)	Jonathan Leung (869)	Law Dominic (647)	Gay Shin Lee (001)	
		Tan Xuan Zhi, Cheri (865)	Woo Jon Hou Ainsley (667)			-					Tan Yih Reng (169)	
4	3	5	5	4	3	3	4	4	4	4	5	1
Wireless	Cloud	Cloud	Wireless	App / video	App / video	Connectivity	App / messaging	Connectivity	Wireless	Connectivity	App / messaging	Connectivity

Suggestions:

- Open-source project (mininet, MCP, openclaw, open5GS, openshift, K8S…)
- New features / new capability / new service / new observation
- Can AI help here? Do we need to design something new to serve an AI-enabled economy?

We will have clusters of groups based on the topics you will focus on (~ 4-5 clusters)

Rubrics

Rubrics

Category	Criteria	Grade	Remarks
Problem statement & literature survey (4pt)	Clear description of the overall problem domain & background, good motivation (i.e., justification of the importance of the problem)	1	
	Well formulated problem statement	1	
	Survey of related work (clear, extensive, up-to-date, in-depth, critical)	2	~ 3 pages in report
Approach formulation and execution (9pt)	Clear description of your approach to tackle the problem	1	
	Good justification of the choice of your approach	2	
	The execution (depending on the nature of your problem, this may include: protocol analysis, code analysis, experiment design, coding, measurements / collection of traces, demo system setup, and more)	6	Document challenges you have overcome. Emphasize ***"do"*** aspect rather than ***"read"***.
Sharing of findings (5pt)	Clear presentation of your main achievements using evidence	2	
	Report write-up & presentation quality	3	
Bonus points	Vote by fellow students	Up to 2pt	
Individual grade adjustment	Each member's working journal and peer-review score will be used to finalize individual's grade		

Individual submission format

- My main contributions to the project include:
  - Design and carry out the …. experiment, together with Alice
  - ……
- My working logs

Index	Activities	Week	Number of hours	Notes
1	Reading reference paper 1	4	2	Presented to the team on ….
2	Setup … tool	5	3	Solved the issues ….
	…			
n	Write up report Section 3	10	4	Together with ….
	Total hours		30	

Peer review table (example)

Teammate	My rating for him/her (0-5)	Notes (examples)
Alice	5	Lead the team in implementing the prototype
Bob	3	Absent in most of our group meetings, but did implement what he is assigned to do
…		